La Scuola d'Italia / Dr. Huson / IB Math
20 May 2026

First & Last name:

8.2 Classwork: Rational Functions — Toward the Maturità

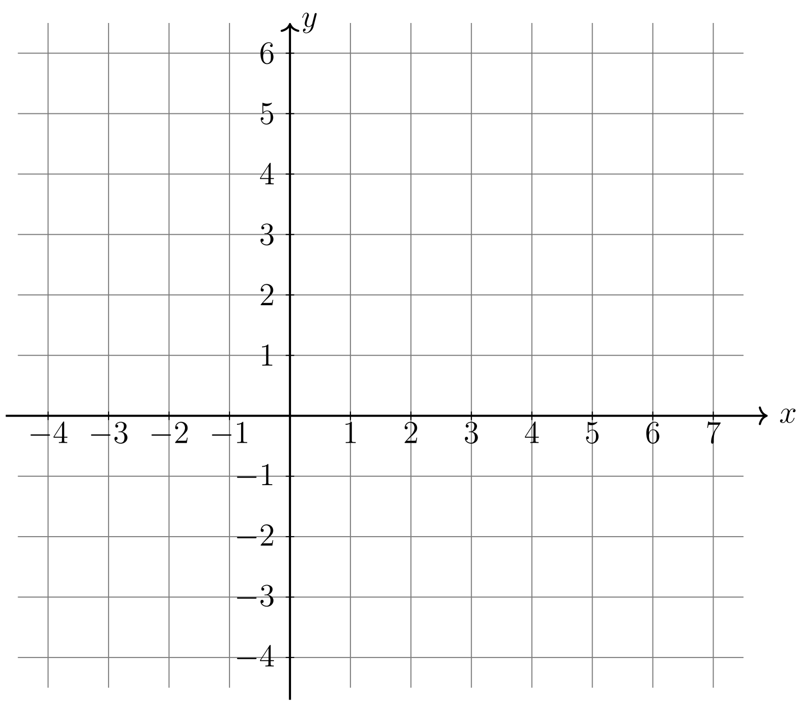
1. Consider $f(x) = \dfrac{x+3}{x-2}$.

   (a) State the equation of the vertical asymptote.

   (b) Verify by long division (or otherwise) that $f(x) = 1 + \dfrac{5}{x-2}$. Use this form to state the equation of the horizontal asymptote.

   (c) Find the x-intercept and the y-intercept.

   (d) Sketch f on the axes below. Mark and label the asymptotes and intercepts.



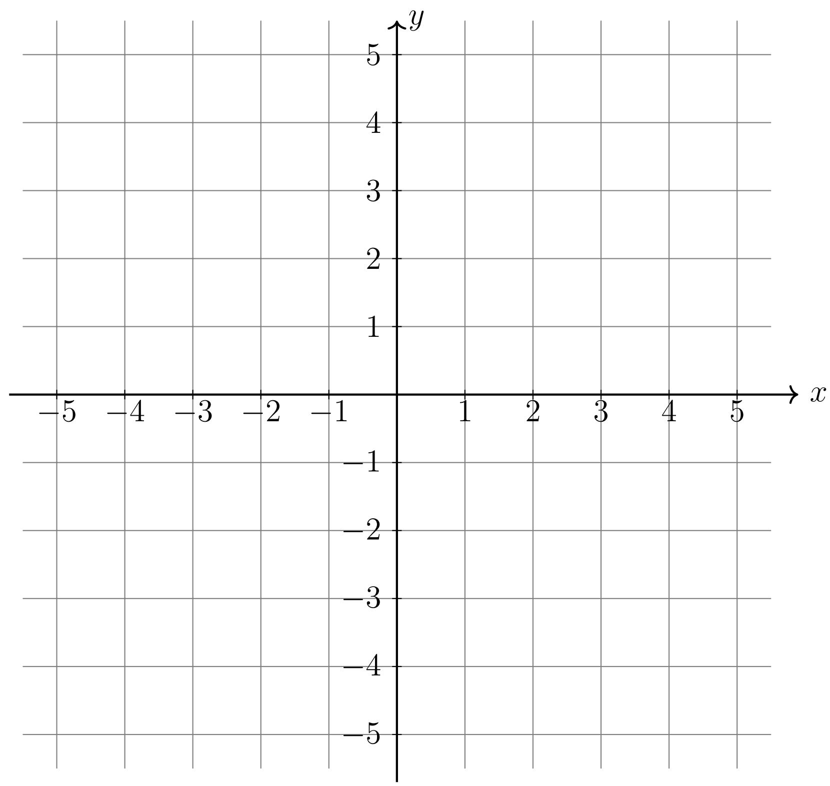
2. Consider the family of functions $f_a(x) = \dfrac{x-2}{x-a}$, with $a \neq 2$.

(a) For an arbitrary value of a, state the equation of the vertical asymptote of f_a.

(b) Find the horizontal asymptote of f_a. Does its equation depend on a?

(c) Compute $f_a(2)$. What does this tell you about a point shared by every curve in the family?

(d) Now set $a = -1$. Find the y-intercept of f_{-1} and sketch its graph on the axes below. Mark the asymptotes (dashed) and the intercepts.



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3. Consider $g(x) = \dfrac{x^2 + x}{x^2 + 1}$.

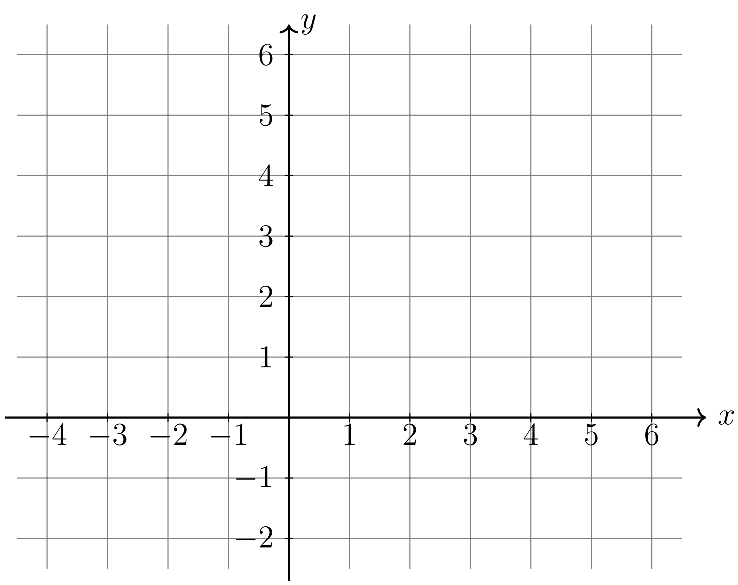
(a) State the domain of g. (Hint: look at the denominator carefully — can it equal zero?)

(b) Compute $\lim\limits_{x \to +\infty} g(x)$ and $\lim\limits_{x \to -\infty} g(x)$. Hence state the horizontal asymptote.

(c) Find the x-intercepts (note: there are two) and the y-intercept.

(d) Complete the table of values, then sketch the graph below.

x	-3	-2	-1	0	1	2	3	5
$g(x)$								



4. Consider the family of functions $f_a(x) = \dfrac{x^2 - ax}{x^2 - a}$, with $a \neq 0$.

   The function g from Problem 3 is the member f_{-1} (check: substituting $a = -1$ gives $g(x) = (x^2 + x)/(x^2 + 1)$).

   (a) Find the domain of f_a in each of the following cases.

      (i) $a < 0$ (for example $a = -1$, $a = -3$):

      (ii) $a > 0$ (for example $a = 4$):

   (b) State the equation of the horizontal asymptote of f_a. (Hint: same idea as Problem 3(b) — compare leading terms.)

   (c) Set $f_a(x)$ equal to the horizontal asymptote you found in (b), and solve for x. Show that the solution does not depend on a (provided $a \neq 1$). Conclude that every curve Ω_a in the family passes through one common point on its horizontal asymptote. State this point.

   (d) Compute $f_a(0)$. Then show, using the quotient rule, that $f_a'(0) = 1$ for every $a \neq 0$. Hence write down the equation of the line tangent to Ω_a at the origin. Is this line the same for every a?

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5. Consider $h(x) = \dfrac{x^3 + 4}{x^2}$.

   (a) State the domain of h, and the equation of the vertical asymptote.

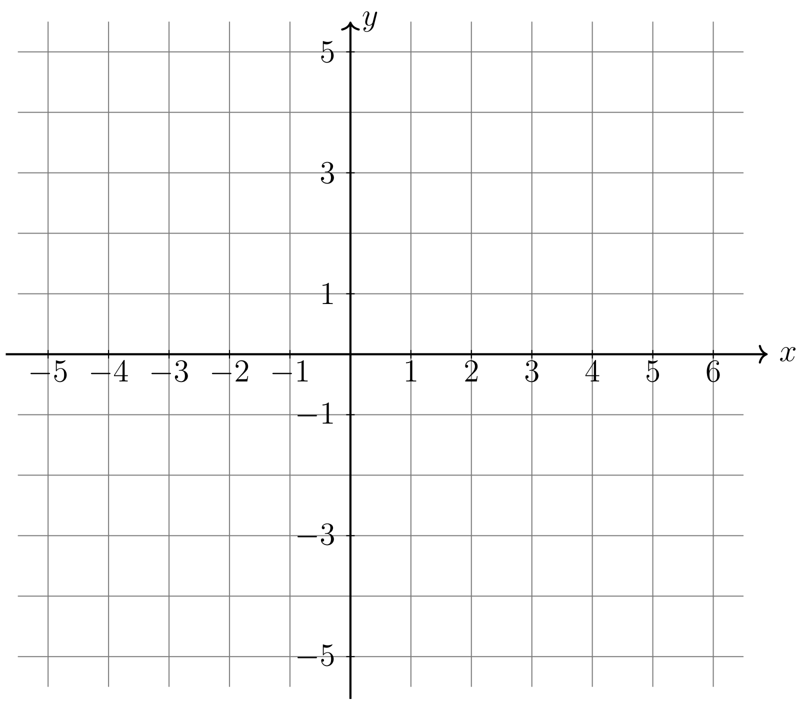
   (b) Show that $h(x) = x + \dfrac{4}{x^2}$.

   (c) Use part (b) to explain why the graph of h approaches the line $y = x$ as $|x|$ grows large. This line is called the *oblique asymptote* of h.

   (d) Complete the table of values for h.

x	-3	-2	-1	-0.5	0.5	1	2	4
$h(x)$								

   (e) Sketch the graph on the axes below. Show both the vertical asymptote and the oblique asymptote $y = x$ as dashed lines.



Problemi della Maturità

(Use separate paper for these problems.)

6. **PROBLEMA 2 — Sessione ordinaria 2023.**

   Fissato un parametro reale a, con $a \neq 0$, si consideri la funzione f_a così definita:

   $$f_a(x) = \frac{x^2 - ax}{x^2 - a}$$

   il cui grafico sarà indicato con Ω_a.

   (a) Al variare del parametro a, determinare il dominio di f_a, studiarne le eventuali discontinuità e scrivere le equazioni di tutti i suoi asintoti.

   (b) Mostrare che, per $a \neq 1$, tutti i grafici Ω_a intersecano il proprio asintoto orizzontale in uno stesso punto e condividono la stessa retta tangente nell'origine.

7. **PROBLEMA 1 — Sessione ordinaria 2024.**

   Si consideri $f_{a,b}(x) = \dfrac{ax^3 + b}{x^2}$, con $a, b \in \mathbb{R}$.

   (a) Determinare i valori dei parametri in modo che la retta t, di equazione $7x+y-12 = 0$, sia tangente al grafico di $f_{a,b}(x)$ nel suo punto P di ascissa $x = 1$.

   Si ponga, d'ora in avanti, $a = 1$ e $b = 4$.

   (b) Studiare la funzione $f(x) = \dfrac{x^3 + 4}{x^2}$ e tracciarne il grafico γ.